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To

Journal of Theoretical Probability,

December 31, 2022

Dear Editor,

In this paper we investigate the conditions required to guarantee the existence of an invariant measure to the stochastic heat equation on the whole space $\mathbb{R}^d$ for $d \geq 3$ with an unbounded Lipschitz diffusion term b . Namely, $\partial_t u(t, x) - 1/2 \Delta_x u(t, x) = b(u(t, x)) \dot{W}$ with the initial condition $u(0, \cdot) = \mu(\cdot)$ and where the Gaussian noise, $\dot{W}$, is white in time and colored in space. The major challenge to the problem is finding conditions which guarantee that the probability moments of the solution are bounded in time and it is often the case that authors require the addition of a drift term to the equation which help control the usual exponential growth of the moments in time [5, 6, 7]. We do not consider such a drift term and the literature for such a set up is far more scarce than when considering the added drift term.

Our paper is inspired by [1] where they prove existence of an invariant measure under the same setting of our paper, ie on the whole space and with an unbounded and Lipschitz diffusion term, when the initial condition is the constant 1. However, their techniques essentially allow for all bounded functions. Moreover, they impose a hard to verify condition on their noise which often is too complicated to verify for many cases of interest.

We expand on this by verifying the existence of an invariant measure for a wider class of initial conditions which include unbounded functions and measures such as the Dirac delta measure. Moreover, we provide explicit and easy to verify conditions on the noise which allow for the existence of an invariant measure and we give several examples as well. We note that our improvements are due to the fact that we work under the random field framework of Walsh [3] where as in the past, this problem has usually been studied under the theory of the stochastic integral laid out by DaPrato and Zabczyk [4]. While both theories have been shown to be equivalent [2], they both offer their own unique set of tools to work with.

Thank you very much for handling our paper.

Sincerely,

Le Chen and Nicholas Eisenberg

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